

Ex 5.2.

P2. Solution:

$$\det A = 1 \cdot \det \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} - 1 \cdot \det \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$= -1 - 1$$

$$= -2$$

$$\det B = 0$$

$$\det C = \det A \cdot \det B = 0$$

P4. Solution:

(a) Rows 3, 4 & 5 are linearly dependent since their 1st, 2nd, 3rd components are all zero.

(b). There is ^{always} at least one entry that is zero when choose one element from rows 3, 4 & 5 with different permutations.

P2. Solution:

$$C = \begin{bmatrix} 6 & -3 \\ -1 & 2 \end{bmatrix}$$

$$\det B = 7 \cdot \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix}$$

$$= 7 \cdot (12 - 15) = -21$$

P13. Solution:

$$C = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

$$AC^T = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

P15. Solution:

$$(a) C_1 = 0, C_2 = -1, C_3 = 0, C_4 = 1$$

$$(b) C_n = -C_{n-2} \Rightarrow C_{10} = -C_8 = C_6 = -C_4 = -1$$

P17 Solution:

$$(a) \begin{vmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} - \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} - 111$$

$$\Rightarrow E_n = E_{n-1} - E_{n-2}$$

$$(a) E_1 = 1, E_2 = 0, E_3 = -1, E_4 = -1$$

$$E_5 = 0, E_6 = 1, E_7 = 1, E_8 = 0$$

P18. Solution:

$$F_4 = \begin{vmatrix} 1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix} = F_3 + F_2$$

$$\Rightarrow F_n = F_{n-1} + F_{n-2}$$

P20. Solution:

Using the big formula, each term only contain one item from row 4.

$$x = \underline{a}, \underline{b}, \text{ and } \underline{c}$$

P27. Solution:

$$\begin{bmatrix} I & \\ -CA^T & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & D - CA^TB \end{bmatrix}$$

$$\det \begin{bmatrix} I & \\ -CA^T & I \end{bmatrix} \det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det \begin{bmatrix} A & B \\ 0 & D - CA^TB \end{bmatrix}$$

$$1 \cdot \begin{vmatrix} A & B \\ C & D \end{vmatrix} = \begin{vmatrix} A & B \\ 0 & D - CA^TB \end{vmatrix}$$

$$\begin{vmatrix} A & B \\ C & D \end{vmatrix} = |A| \cdot |D - CA^TB| = |AD - ACA^TB|$$

$$\text{if } AC = CA \\ = |AD - BC|$$

§5.3

P2. Solution

$$(a) A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\det A = ad - bc$$

$$|B_1| = \begin{vmatrix} 1 & b \\ 0 & d \end{vmatrix} = d$$

$$|B_2| = \begin{vmatrix} a & 1 \\ c & 0 \end{vmatrix} = -c$$

$$\Rightarrow x = \frac{d}{ad-bc}, y = -\frac{c}{ad-bc}$$

$$(b) A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$|B_2| = \begin{vmatrix} a & 1 & c \\ d & 0 & f \\ g & 0 & i \end{vmatrix} = -\begin{vmatrix} d & f \\ g & i \end{vmatrix} = -di + gf$$

$$\Rightarrow y = \frac{-di + gf}{\det A}$$

P6. solution:

$$(a) \det A = 3$$

$$C = \begin{bmatrix} 3 & 0 & 0 \\ -2 & 1 & -4 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{C^T}{\det A} = \begin{bmatrix} 1 & -2/3 & 0 \\ 0 & 1/3 & 0 \\ 0 & -4/3 & 1 \end{bmatrix}$$

$$(b) \det A = 4$$

$$C = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{C^T}{\det A} = \begin{bmatrix} 3/4 & 1/2 & 1/4 \\ 1/2 & 1 & 1/2 \\ 1/4 & 1/2 & 3/4 \end{bmatrix}$$

P8. solution:

$$C = \begin{bmatrix} 6 & -3 & 0 \\ 3 & 1 & -1 \\ -6 & 2 & 1 \end{bmatrix} \quad AC^T = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\det A = 3$$

P9. solution:

$$A = (C^T)^{-1} \quad \det C^T = \det C =$$

P11. solution:

$$A^{-1} = \frac{C^T}{\det A} = \pm C^T$$

Each entry of C^T is the determinant of a matrix consisting of integers, so that it is an integer.

P18. solution:

$$|V| = \begin{vmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{vmatrix} = 3 \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} - \begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 1 & 1 \end{vmatrix}$$

$$= 24 - 2 + (-2) = 20$$

$$u_1 = (3, 1, 1), u_2 = (1, 3, 1), u_3 = (1, 1, 3)$$

$$u_1 \times u_2 = \begin{vmatrix} i & j & k \\ 3 & 1 & 1 \\ 1 & 3 & 1 \end{vmatrix} = -2i - 2j + 8k \Rightarrow \|u_1 \times u_2\| = 6\sqrt{2}$$

$$u_1 \times u_3 = \begin{vmatrix} i & j & k \\ 3 & 1 & 1 \\ 1 & 1 & 3 \end{vmatrix} = 2i - 8j + 2k \Rightarrow \|u_1 \times u_3\| = 6\sqrt{2}$$

$$u_2 \times u_3 = \begin{vmatrix} i & j & k \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{vmatrix} = 8i - 2j - 2k \Rightarrow \|u_2 \times u_3\| = 6\sqrt{2}$$

P24.

$$A = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \Rightarrow A^T A = \begin{bmatrix} a^T a & 0 & 0 \\ 0 & b^T b & 0 \\ 0 & 0 & c^T c \end{bmatrix}$$

$$\det A = \|a\|^2 \|b\|^2 \|c\|^2$$

$$\Rightarrow \det A = \pm \sqrt{\|a\| \|b\| \|c\|}$$

P27. Solution

$$\frac{1}{6}$$

$$\frac{1}{24}$$

P32. Solution:

$$u = (2, 4, 0), v = (-1, 3, 0)$$

$$w = (1, 2, 2)$$

$$u \times v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 4 & 0 \\ -1 & 3 & 0 \end{vmatrix} = 10\hat{k}$$

$$\Rightarrow \text{Base area} = \|u \times v\| = 10$$

$$(u \times v) \cdot w = (0, 0, 10) \cdot (1, 2, 2) \\ = 20.$$

$$\Rightarrow \text{Volume} = 20$$

$$\Rightarrow \text{Height} = \frac{\text{Volume}}{\|u \times v\|} = 2.$$